1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

Primary hypothesis: In HER2-positive breast cancer, elevated MKI67 mRNA expression (RNA-seq_MKI67_signal) is independently associated with increased tumor cell proliferation and shorter overall survival after adjusting for standard clinicopathologic covariates; this prognostic association is modified by trastuzumab therapy such that the strength and direction of the MKI67–OS relationship differ by trastuzumab exposure. Secondary hypotheses: (1) MKI67 mRNA expression correlates with the PAM50 proliferation score and with lower immune infiltration, and a combined MKI67 + PAM50 proliferation score provides superior prognostic discrimination versus either marker alone; (2) The MKI67–OS association is reproducible in external HER2-positive cohorts and correlates with IHC Ki-67 on matched samples, supporting clinical transferability; (3) In preclinical models, modulation of MKI67 expression alters proliferation and impacts sensitivity to HER2-targeted therapies, suggesting a functional role in therapy response and potential as a therapeutic biomarker.

2. Why This Gene — Not Others

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions